Lesson 9.6 - Chebyshev's Theorem

> **Video:** Chebyshev's Theorem by The Organic Chemistry Tutor **Chapter:** 9 - Confidence Intervals

Statement

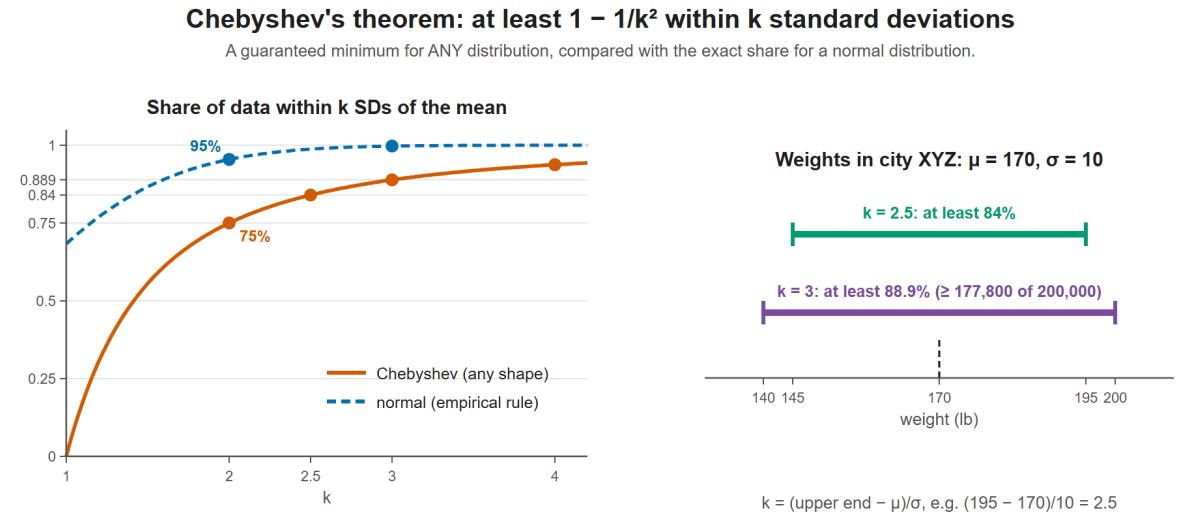
For **any** distribution, whatever its shape, the proportion of data within k standard deviations of the mean is **at least**

$$1 - \frac{1}{k^2}, \qquad k > 1$$

- It gives a **minimum**, not an exact value. The true proportion can be higher.
- It works for **all** distributions, not only normal ones.
- In the video the formula is spoken as "1 over 1 minus k squared". The correct form is $1 - 1/k^2$.

Chebyshev vs the empirical rule

Within	Empirical rule (normal only)	Chebyshev (any distribution, minimum)
1σ	68%	— (k must be > 1)
2σ	95%	$1 - \frac{1}{4} = 75\%$
3σ	99.7%	$1 - \frac{1}{9} \approx 88.9\%$
4σ	≈ 99.99%	$1 - \frac{1}{16} = 93.75\%$



The orange curve is always below the blue one: Chebyshev is a cautious lower bound.

Left: Chebyshev's guaranteed minimum (orange) is always below the exact share for normal data (blue). For example, it guarantees 75% within 2 SDs, while a normal distribution has 95%. Right: the two intervals from the weight example.

Finding k

If an interval is centered on the mean, $\mu \pm k\sigma$, then

$$k = \frac{x - \mu}{\sigma}$$

using the upper end x (the lower end gives the same k).

Example - weights in city XYZ

Mean weight $\mu = 170$ lb, $\sigma = 10$ lb, distribution shape unknown.

(a) Minimum percentage of people weighing between 145 and 195 lb

Both ends are 25 lb from the mean, so

$$k = \frac{195 - 170}{10} = 2.5$$

$$1 - \frac{1}{2.5^2} = 1 - \frac{1}{6.25} = 0.84$$

At least 84% of residents weigh between 145 and 195 lb. The real figure could be 86%, 89%, 95%, ...

(b) The city has 200,000 residents. What is the minimum number weighing between 140 and 200 lb?

Both ends are 30 lb from the mean, so

$$k = \frac{200 - 170}{10} = 3, \qquad 1 - \frac{1}{9} = \frac{8}{9} \approx 0.889$$

$$\frac{8}{9} \times 200{,}000 \approx 177{,}778 \text{ people}$$

(The video rounds to 0.889 first and gets 177,800.)

So $P(140 \le X \le 200) \ge 0.889$, and **at least about 177,800** residents weigh between 140 and 200 lb.

Key takeaways

- Chebyshev: at least $1 - 1/k^2$ of any data set lies within k SDs of the mean ($k > 1$).

- Use it when the distribution is **unknown or not normal**. Use the empirical rule only for normal data.
- $k = (x - \mu)/\sigma$ for an interval centered on the mean.
- Answers are **lower bounds**: say "at least".